

Hriday Adani

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EDUCATION

Rutgers University - New Brunswick

New Brunswick, NJ

Bachelor's of Science; Major in Computer Science and Economics

Expected Graduation - May 2027

Relevant Coursework : Data Structures, Artificial Intelligence, Algorithms

PROJECTS

EpochOS - AI-Native Local System Layer | *ReactJS, Tailwind CSS, Tauri CLI, Node Js*

- Designed a Local-First AI-Native system layer augments traditional OS interactions with autonomous workflows
- Built a desktop shell using Tauri, React, and Tailwind, achieving a 90% smaller build size compared to similar apps
- Integrated OpenAI and local LLMs via Ollama to enable offline task automation, improving efficiency by 40%
- Devised a memory engine using ChromaDB with OpenAI embeddings and RAG, raising retrieval speed by 65%
- Employed system-level APIs (file I/O) to automate 15+ repetitive workflows, increasing manual task efficiency
- Managed a local storage system for handling large volumes of user data and states, supporting 1 TB data capacity

ClauseWatch - AI-Powered Contract Intelligence SaaS | *Next.js, FastAPI, PostgreSQL, OpenAI*

- Built an LLM-powered contract extraction pipeline with per-field confidence scoring across PDF/DOCX formats
- Engineered 5 independently-secured HMAC token systems including webhooks, downloads, password reset, email verification, and cron auth, each with a dedicated secret and constant-time comparison to prevent timing attacks.
- Designed a multi-tenant data isolation architecture by enforcing tenant-scoped queries at the ORM layer across every API endpoint, verified via 6 automated integration tests confirming zero cross-tenant data leakage
- Evolved a production PostgreSQL schema through 9 zero-downtime Alembic migrations, including safe handling of Postgres enum-type constraints unsupported by migration autogeneration, applied in an isolated transaction.

AccessForge - Accessibility Remediation Pipeline | *FastAPI, LangGraph, PostgreSQL, Reids/RQ, Docker*

- Developed an end-to-end accessibility remediation platform that scans React apps, generates and independently verifies WCAG fixes, and opens GitHub PRs, designed as a deterministic state machine using LangGraph
- Architected a crash-recoverable async pipeline using PostgreSQL-backed state and Redis/RQ workers, verified via a real worker crash captured during development, resuming task with 0 duplicate GitHub branches or lost progress
- Achieved 100% build-pass, first-attempt-fix, and regression free rates on fully verified LLM-assisted patches across all 20-case accessibility benchmarks with 6 WCAG categories through multimodal visual-regression comparison
- Eliminated a confirmed class of false-positive LLM-assisted fixes by buiding a confidence-gated source localization check that defers to human review instead instead of guessing when no reliable signal matches any candidate file
- Integrated ElevenLabs TTS API with serial processing, delivering low latency audio summaries in 10+ languages
- Deployed secure authentication using Google OAuth and Cloud Storage with encryption for document handling

WORK EXPERIENCES

Scientific Software Research Assistant

Jul. 2026 - Present

RCSB Protein Data Bank

New Brunswick, NJ

- Orchestrated a new MCP server tool exposing PDB search API to LLM agents enabling multi-service AND/OR queries in a single call, by composing the filter nodes from the 7 extended tools through a custom validation layer
- Rectified 2 silent data corruption bugs in MCP server's result-grouping logic by tracing parameter flow against against the tool's own docstrings, preventing the formation of invalid API parameters and user configurations

Summer Research Fellow (CSI SURF)

May. 2026 - Jul. 2026

Rutgers Artificial Intelligence and Data Science Collaboratory

New Brunswick, NJ

- Redesigned a dictionary-driven column-typing model in py-mmCIF's data compression pipeline with an automatic detection and value-scanning classifier, reducing molecular structure file size by 27%, eliminating the dependencies
- Collaborated with 4 researchers to research on binary encoding chain algorithms for data compression and reduced compressed structure file sizes by up to 76% across 37 varied PDB structures by optimizing floating-point encoding

TECHNICAL SKILLS

Languages: Python, Java, C, PHP, JavaScript (React, Next), SQL, HTML/CSS, Rust

Libraries/Frameworks: Tailwind CSS, Tuari, Flask, FastAPI, SciKit-Learn, Pandas

Developer Tools: VS Code, IntelliJ, Git, Docker, WSL, Alembic, Neon, Cloudflare, MCP